AWS IAM Project

Full commands with a simple explanation for each

IMPORTANT: This project is not subject to actual execution

This is a scenario for learning only and is NOT meant to be run for real. All commands, names and the account ID (123456789101) are sample values. Do not run them on a real AWS account.

What this project does

- Creates 5 groups and 5 users, and puts each user in a group.
- For each group, creates a bad policy (too many permissions), checks it, then removes it.
- Then creates the correct policy (only the needed permissions) and attaches it.
- Red headings are bad policies. Green headings are correct policies.

Summary

Group	User	Bad policy (deleted)	Correct policy
IAMW-Developers	ahmed.dev	developers-bad-readonly	developers-readonly
IAMW-Security	sara.sec	security-audit-bad	security-audit
IAMW-Finance	mona.fin	finance-bad-readonly	finance-readonly
IAMW-Operations	omar.ops	operations-bad-policy	operations-policy
IAMW-Auditors	ali.audit	auditor-bad-policy	auditor-policy

1) Create the groups

One group for each team.

1.

```
aws iam create-group --group-name IAMW-Developers
```

Creates the group IAMW-Developers for the developers team.

2.

```
aws iam create-group --group-name IAMW-Operations
```

Creates the group IAMW-Operations for the operations team.

3.

```
aws iam create-group --group-name IAMW-Security
```

Creates the group IAMW-Security for the security team.

4.

```
aws iam create-group --group-name IAMW-Finance
```

Creates the group IAMW-Finance for the finance team.

5.

```
aws iam create-group --group-name IAMW-Auditors
```

Creates the group IAMW-Auditors for the auditors team.

2) Create the users

One user for each team.

6.

```
aws iam create-user --user-name ahmed.dev
```

Creates the user ahmed.dev.

7.

```
aws iam create-user --user-name sara.sec
```

Creates the user sara.sec.

8.

```
aws iam create-user --user-name omar.ops
```

Creates the user omar.ops.

9.

```
aws iam create-user --user-name mona.fin
```

Creates the user mona.fin.

10.

```
aws iam create-user --user-name ali.audit
```

Creates the user ali.audit.

3) Add each user to a group

The user gets the permissions of the group.

11.

```
aws iam add-user-to-group --user-name ahmed.dev --group-name IAMW-Developers
```

Adds ahmed.dev to the group IAMW-Developers.

12.

```
aws iam add-user-to-group --user-name sara.sec --group-name IAMW-Security
```

Adds sara.sec to the group IAMW-Security.

13.

```
aws iam add-user-to-group --user-name omar.ops --group-name IAMW-Operations
```

Adds omar.ops to the group IAMW-Operations.

14.

```
aws iam add-user-to-group --user-name mona.fin --group-name IAMW-Finance
```

Adds mona.fin to the group IAMW-Finance.

15.

```
aws iam add-user-to-group --user-name ali.audit --group-name IAMW-Auditors
```

Adds ali.audit to the group IAMW-Auditors.

4) Check the users and groups

These commands only show information.

16.

```
aws iam get-user --user-name ahmed.dev
```

Shows the details of the user ahmed.dev.

17.

```
aws iam get-user --user-name sara.sec
```

Shows the details of the user sara.sec.

18.

```
aws iam get-user --user-name omar.ops
```

Shows the details of the user omar.ops.

19.

```
aws iam get-user --user-name mona.fin
```

Shows the details of the user mona.fin.

20.

```
aws iam get-user --user-name ali.audit
```

Shows the details of the user ali.audit.

21.

```
aws iam get-group --group-name IAMW-Developers
```

Shows the group IAMW-Developers and its members.

22.

```
aws iam get-group --group-name IAMW-Security
```

Shows the group IAMW-Security and its members.

23.

```
aws iam get-group --group-name IAMW-Operations
```

Shows the group IAMW-Operations and its members.

24.

```
aws iam get-group --group-name IAMW-Finance
```

Shows the group IAMW-Finance and its members.

25.

```
aws iam get-group --group-name IAMW-Auditors
```

Shows the group IAMW-Auditors and its members.

26.

```
aws iam list-users
```

Lists all users in the account.

27.

```
aws iam list-groups
```

Lists all groups in the account.

5) Prepare the project folder

A folder to keep the policy files.

28.

```
mkdir -p iam-project
```

Creates the folder iam-project.

29.

```
cd iam-project
```

Goes into the folder.

6-A) Bad developers policy: developers-bad-readonly

30.

```
touch developers-bad-readonly-policy.json
```

Creates an empty file for the policy.

31.

```
nano developers-bad-readonly-policy.json
```

Opens the file to write the policy. Save with Ctrl+O then Enter, and exit with Ctrl+X.

32.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "*",
      "Resource": "*"
    }
  ]
}
```

This is the content of the file. This policy is bad: it allows every action on everything.

33.

```
aws iam create-policy --policy-name developers-bad-readonly --policy-document
file://developers-bad-readonly-policy.json
```

Creates the policy developers-bad-readonly from the JSON file.

34.

```
aws iam attach-group-policy --group-name IAMW-Developers --policy-arn
arn:aws:iam::123456789101:policy/developers-bad-readonly
```

Attaches the policy to the group IAMW-Developers.

35.

```
aws iam list-attached-group-policies --group-name IAMW-Developers
```

Shows the policies attached to IAMW-Developers.

36.

```
aws iam get-policy --policy-arn arn:aws:iam::123456789101:policy/developers-bad-readonly
```

Shows the details of the policy.

37.

```
aws iam get-policy-version --policy-arn arn:aws:iam::123456789101:policy/developers-bad-readonly
--version-id v1
```

Shows the permissions inside version v1 of the policy.

38.

```
aws iam detach-group-policy --group-name IAMW-Developers --policy-arn
arn:aws:iam::123456789101:policy/developers-bad-readonly
```

Removes the bad policy from the group.

39.

```
aws iam delete-policy --policy-arn arn:aws:iam::123456789101:policy/developers-bad-readonly
```

Deletes the bad policy.

6-B) Correct developers policy: developers-readonly

40.

```
touch developers-readonly-policy.json
```

Creates an empty file for the policy.

41.

```
nano developers-readonly-policy.json
```

Opens the file to write the policy. Save with Ctrl+O then Enter, and exit with Ctrl+X.

42.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "ec2:DescribeInstances",
        "ec2:DescribeVolumes",
        "ec2:DescribeSecurityGroups"
      ],
      "Resource": "*"
    }
  ]
}
```

This is the content of the file. This policy is good: it only allows viewing EC2 information.

43.

```
aws iam create-policy --policy-name developers-readonly --policy-document
file://developers-readonly-policy.json
```

Creates the policy developers-readonly from the JSON file.

44.

```
aws iam attach-group-policy --group-name IAMW-Developers --policy-arn
arn:aws:iam::123456789101:policy/developers-readonly
```

Attaches the policy to the group IAMW-Developers.

45.

```
aws iam list-attached-group-policies --group-name IAMW-Developers
```

Shows the policies attached to IAMW-Developers.

46.

```
aws iam get-policy --policy-arn arn:aws:iam::123456789101:policy/developers-readonly
```

Shows the details of the policy.

47.

```
aws iam get-policy-version --policy-arn arn:aws:iam::123456789101:policy/developers-readonly
--version-id v1
```

Shows the permissions inside version v1 of the policy.

7-A) Bad security policy: security-audit-bad

48.

```
touch security-audit-bad-policy.json
```

Creates an empty file for the policy.

49.

```
nano security-audit-bad-policy.json
```

Opens the file to write the policy. Save with Ctrl+O then Enter, and exit with Ctrl+X.

50.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "iam:CreateUser",
        "iam:DeleteUser",
        "iam:UpdateUser"
      ],
      "Resource": "*"
    }
  ]
}
```

This is the content of the file. This policy is bad: it lets the team create, delete and update users.

51.

```
aws iam create-policy --policy-name security-audit-bad --policy-document
file://security-audit-bad-policy.json
```

Creates the policy security-audit-bad from the JSON file.

52.

```
aws iam attach-group-policy --group-name IAMW-Security --policy-arn
arn:aws:iam::123456789101:policy/security-audit-bad
```

Attaches the policy to the group IAMW-Security.

53.

```
aws iam list-attached-group-policies --group-name IAMW-Security
```

Shows the policies attached to IAMW-Security.

54.

```
aws iam get-policy --policy-arn arn:aws:iam::123456789101:policy/security-audit-bad
```

Shows the details of the policy.

55.

```
aws iam get-policy-version --policy-arn arn:aws:iam::123456789101:policy/security-audit-bad --version-id
v1
```

Shows the permissions inside version v1 of the policy.

56.

```
aws iam detach-group-policy --group-name IAMW-Security --policy-arn
arn:aws:iam::123456789101:policy/security-audit-bad
```

Removes the bad policy from the group.

57.

```
aws iam delete-policy --policy-arn arn:aws:iam::123456789101:policy/security-audit-bad
```

Deletes the bad policy.

7-B) Correct security policy: security-audit

58.

```
touch security-audit-policy.json
```

Creates an empty file for the policy.

59.

```
nano security-audit-policy.json
```

Opens the file to write the policy. Save with Ctrl+O then Enter, and exit with Ctrl+X.

60.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "iam:ListUsers",
        "iam:ListGroups",
        "iam:ListRoles",
        "iam:ListPolicies",
        "iam:ListAttachedUserPolicies",
        "iam:ListAttachedGroupPolicies",
        "iam:ListAttachedRolePolicies",
        "iam:GetPolicy",
        "iam:GetPolicyVersion"
      ],
      "Resource": "*"
    }
  ]
}
```

This is the content of the file. This policy is good: it only allows viewing IAM information.

61.

```
aws iam create-policy --policy-name security-audit --policy-document file://security-audit-policy.json
```

Creates the policy security-audit from the JSON file.

62.

```
aws iam attach-group-policy --group-name IAMW-Security --policy-arn
arn:aws:iam::123456789101:policy/security-audit
```

Attaches the policy to the group IAMW-Security.

63.

```
aws iam list-attached-group-policies --group-name IAMW-Security
```

Shows the policies attached to IAMW-Security.

64.

```
aws iam get-policy --policy-arn arn:aws:iam::123456789101:policy/security-audit
```

Shows the details of the policy.

65.

```
aws iam get-policy-version --policy-arn arn:aws:iam::123456789101:policy/security-audit --version-id v1
```

Shows the permissions inside version v1 of the policy.

8-A) Bad finance policy: finance-bad-readonly

66.

```
touch finance-readonly-bad-policy.json
```

Creates an empty file for the policy.

67.

```
nano finance-readonly-bad-policy.json
```

Opens the file to write the policy. Save with Ctrl+O then Enter, and exit with Ctrl+X.

68.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "ce:GetCostAndUsage",
        "ce:GetCostForecast",
        "budgets:ViewBudget",
        "ce:GetDimensionValues",
        "ce:ListCostCategoryDefinitions",
        "budgets:DescribeBudget",
        "aws-portal:ViewUsage",
        "aws-portal:ViewBilling"
      ],
      "Resource": "*"
    }
  ]
}
```

This is the content of the file. This policy is bad: it gives more billing permissions than needed.

69.

```
aws iam create-policy --policy-name finance-bad-readonly --policy-document
file://finance-readonly-bad-policy.json
```

Creates the policy finance-bad-readonly from the JSON file.

70.

```
aws iam attach-group-policy --group-name IAMW-Finance --policy-arn
arn:aws:iam::123456789101:policy/finance-bad-readonly
```

Attaches the policy to the group IAMW-Finance.

71.

```
aws iam list-attached-group-policies --group-name IAMW-Finance
```

Shows the policies attached to IAMW-Finance.

72.

```
aws iam get-policy --policy-arn arn:aws:iam::123456789101:policy/finance-bad-readonly
```

Shows the details of the policy.

73.

```
aws iam get-policy-version --policy-arn arn:aws:iam::123456789101:policy/finance-bad-readonly
--version-id v1
```

Shows the permissions inside version v1 of the policy.

74.

```
aws iam detach-group-policy --group-name IAMW-Finance --policy-arn
arn:aws:iam::123456789101:policy/finance-bad-readonly
```

Removes the bad policy from the group.

75.

```
aws iam delete-policy --policy-arn arn:aws:iam::123456789101:policy/finance-bad-readonly
```

Deletes the bad policy.

8-B) Correct finance policy: finance-readonly

76.

```
touch finance-readonly.json
```

Creates an empty file for the policy.

77.

```
nano finance-readonly.json
```

Opens the file to write the policy. Save with Ctrl+O then Enter, and exit with Ctrl+X.

78.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "ce:GetCostAndUsage",
        "budgets:ViewBudget"
      ],
      "Resource": "*"
    }
  ]
}
```

This is the content of the file. This policy is good: it only allows viewing costs and budgets.

79.

```
aws iam create-policy --policy-name finance-readonly --policy-document file://finance-readonly.json
```

Creates the policy finance-readonly from the JSON file.

80.

```
aws iam attach-group-policy --group-name IAMW-Finance --policy-arn
arn:aws:iam::123456789101:policy/finance-readonly
```

Attaches the policy to the group IAMW-Finance.

81.

```
aws iam list-attached-group-policies --group-name IAMW-Finance
```

Shows the policies attached to IAMW-Finance.

82.

```
aws iam get-policy --policy-arn arn:aws:iam::123456789101:policy/finance-readonly
```

Shows the details of the policy.

83.

```
aws iam get-policy-version --policy-arn arn:aws:iam::123456789101:policy/finance-readonly --version-id v1
```

Shows the permissions inside version v1 of the policy.

9-A) Bad operations policy: operations-bad-policy

84.

```
touch operations-bad-policy.json
```

Creates an empty file for the policy.

85.

```
nano operations-bad-policy.json
```

Opens the file to write the policy. Save with Ctrl+O then Enter, and exit with Ctrl+X.

86.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "ec2:StartInstances",
        "ec2:StopInstances",
        "ec2:TerminateInstances"
      ],
      "Resource": "*"
    }
  ]
}
```

This is the content of the file. This policy is bad: it lets the team start, stop and delete servers.

87.

```
aws iam create-policy --policy-name operations-bad-policy --policy-document
file://operations-bad-policy.json
```

Creates the policy operations-bad-policy from the JSON file.

88.

```
aws iam attach-group-policy --group-name IAMW-Operations --policy-arn
arn:aws:iam::123456789101:policy/operations-bad-policy
```

Attaches the policy to the group IAMW-Operations.

89.

```
aws iam list-attached-group-policies --group-name IAMW-Operations
```

Shows the policies attached to IAMW-Operations.

90.

```
aws iam get-policy --policy-arn arn:aws:iam::123456789101:policy/operations-bad-policy
```

Shows the details of the policy.

91.

```
aws iam get-policy-version --policy-arn arn:aws:iam::123456789101:policy/operations-bad-policy
--version-id v1
```

Shows the permissions inside version v1 of the policy.

92.

```
aws iam detach-group-policy --group-name IAMW-Operations --policy-arn
arn:aws:iam::123456789101:policy/operations-bad-policy
```

Removes the bad policy from the group.

93.

```
aws iam delete-policy --policy-arn arn:aws:iam::123456789101:policy/operations-bad-policy
```

Deletes the bad policy.

9-B) Correct operations policy: operations-policy

94.

```
touch operations-policy.json
```

Creates an empty file for the policy.

95.

```
nano operations-policy.json
```

Opens the file to write the policy. Save with Ctrl+O then Enter, and exit with Ctrl+X.

96.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ReadOperationalLogs",
      "Effect": "Allow",
      "Action": [
        "logs:FilterLogEvents",
        "logs:DescribeLogGroups",
        "logs:DescribeLogStreams"
      ],
      "Resource": "*"
    },
    {
      "Sid": "ReadCloudWatchMetrics",
      "Effect": "Allow",
      "Action": [
        "cloudwatch:GetMetricData",
        "cloudwatch:ListDashboards"
      ],
      "Resource": "*"
    }
  ]
}
```

This is the content of the file. This policy is good: it only allows reading logs and metrics.

97.

```
aws iam create-policy --policy-name operations-policy --policy-document file://operations-policy.json
```

Creates the policy operations-policy from the JSON file.

98.

```
aws iam attach-group-policy --group-name IAMW-Operations --policy-arn
arn:aws:iam::123456789101:policy/operations-policy
```

Attaches the policy to the group IAMW-Operations.

99.

```
aws iam list-attached-group-policies --group-name IAMW-Operations
```

Shows the policies attached to IAMW-Operations.

100.

```
aws iam get-policy --policy-arn arn:aws:iam::123456789101:policy/operations-policy
```

Shows the details of the policy.

101.

```
aws iam get-policy-version --policy-arn arn:aws:iam::123456789101:policy/operations-policy --version-id
v1
```

Shows the permissions inside version v1 of the policy.

10-A) Bad auditor policy: auditor-bad-policy

102.

```
touch auditor-bad-policy.json
```

Creates an empty file for the policy.

103.

```
nano auditor-bad-policy.json
```

Opens the file to write the policy. Save with Ctrl+O then Enter, and exit with Ctrl+X.

104.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:PutObject",
        "s3:GetObject"
      ],
      "Resource": "*"
    }
  ]
}
```

This is the content of the file. This policy is bad: it lets the auditor read and write S3 files.

105.

```
aws iam create-policy --policy-name auditor-bad-policy --policy-document file://auditor-bad-policy.json
```

Creates the policy auditor-bad-policy from the JSON file.

106.

```
aws iam attach-group-policy --group-name IAMW-Auditors --policy-arn
arn:aws:iam::123456789101:policy/auditor-bad-policy
```

Attaches the policy to the group IAMW-Auditors.

107.

```
aws iam list-attached-group-policies --group-name IAMW-Auditors
```

Shows the policies attached to IAMW-Auditors.

108.

```
aws iam get-policy --policy-arn arn:aws:iam::123456789101:policy/auditor-bad-policy
```

Shows the details of the policy.

109.

```
aws iam get-policy-version --policy-arn arn:aws:iam::123456789101:policy/auditor-bad-policy --version-id
v1
```

Shows the permissions inside version v1 of the policy.

110.

```
aws iam detach-group-policy --group-name IAMW-Auditors --policy-arn
arn:aws:iam::123456789101:policy/auditor-bad-policy
```

Removes the bad policy from the group.

111.

```
aws iam delete-policy --policy-arn arn:aws:iam::123456789101:policy/auditor-bad-policy
```

Deletes the bad policy.

10-B) Correct auditor policy: auditor-policy

112.

```
touch auditor-policy.json
```

Creates an empty file for the policy.

113.

```
nano auditor-policy.json
```

Opens the file to write the policy. Save with Ctrl+O then Enter, and exit with Ctrl+X.

114.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "IAMAccessReview",
      "Effect": "Allow",
      "Action": [
        "iam:ListRoles",
        "iam:ListUsers",
        "iam:ListGroups",
        "iam:ListPolicies",
        "iam:ListAttachedGroupPolicies",
        "iam:ListAttachedUserPolicies",
        "iam:ListAttachedRolePolicies",
        "iam:GetAccountAuthorizationDetails"
      ],
      "Resource": "*"
    },
    {
      "Sid": "AuditEvents",
      "Effect": "Allow",
      "Action": [
        "cloudtrail:LookupEvents"
      ],
      "Resource": "*"
    }
  ]
}
```

This is the content of the file. This policy is good: it only allows reviewing IAM and CloudTrail events.

115.

```
aws iam create-policy --policy-name auditor-policy --policy-document file://auditor-policy.json
```

Creates the policy auditor-policy from the JSON file.

116.

```
aws iam attach-group-policy --group-name IAMW-Auditors --policy-arn
arn:aws:iam::123456789101:policy/auditor-policy
```

Attaches the policy to the group IAMW-Auditors.

117.

```
aws iam list-attached-group-policies --group-name IAMW-Auditors
```

Shows the policies attached to IAMW-Auditors.

118.

```
aws iam get-policy --policy-arn arn:aws:iam::123456789101:policy/auditor-policy
```

Shows the details of the policy.

119.

```
aws iam get-policy-version --policy-arn arn:aws:iam::123456789101:policy/auditor-policy --version-id v1
```

Shows the permissions inside version v1 of the policy.